Notes on Qubits and Single-Qubit Gates

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1 Definitions and basic properties

The qubit

A **qubit** is a two-level quantum system, described by a two-dimensional Hilbert space. In this space one chooses a **computational basis**, which is basis of two orthogonal kets labelled with '0' and '1'.

$$|0\rangle \equiv \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad |1\rangle \equiv \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{1-1}$$

- The states of the computational basis are orthogonal and normalized (they are *orthonormal*):

$$\langle 0|1\rangle = \langle 1|0\rangle = 0 \tag{1-2}$$
$$\langle 0|0\rangle = \langle 1|1\rangle = 1 \tag{1-3}$$

- The generic state of the qubit is written as:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle \tag{1-4}$$

In the general case $\alpha, \beta \in \mathbb{C}$. As global phase terms on the quantum state are insignificant, most often α can be taken as real. The normalization condition (1-4) implies:

$$1 = \langle\psi|\psi\rangle = (\langle 0|\,\alpha^* + \langle 1|\,\beta^*)(\alpha\,|0\rangle + \beta\,|1\rangle) \tag{1-5}$$
$$1 = \alpha^*\alpha + \beta^*\beta = |\alpha|^2 + |\beta|^2 \tag{1-6}$$

- The coefficients α and β are conveniently parametrized by two angles θ and ϕ as:

$$\alpha = \cos\frac{\theta}{2} \qquad \beta = e^{\iota\phi}\sin\frac{\theta}{2} \tag{1-7}$$

with $\phi \in [0, 2\pi)$ and $\theta \in [0, \pi]$. The generic state of the qubit thus reads:

$$|\psi\rangle = \cos\frac{\theta}{2}\,|0\rangle + e^{\iota\phi}\sin\frac{\theta}{2}\,|1\rangle \tag{1-8}$$

This parametrization sets a biunivocal correspondence between the quantum state of the qubit and a point on a spherical surface of unit radius (the **Bloch sphere**, see Fig. 1). The cartesian coordinates of such a point are:

$$\begin{cases} x = \sin\theta\,\cos\phi \\ y = \sin\theta\,\sin\phi \\ z = \cos\theta \end{cases} \tag{1-9}$$

The states $|0\rangle$ and $|1\rangle$ corresponds to the opposite 'poles' of the sphere.

- The point on the sphere is, as well, associated to a unit vector $\vec{u} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Note that orthogonal kets in the Hilbert space are not associated to orthogonal vectors, but to **opposite vectors** in the Bloch sphere representation.

Proof. We take $|\psi\rangle$ as in Eq. (1-8) and we look for an orthogonal state $|\psi'\rangle = \cos\frac{\theta'}{2}\,|0\rangle + e^{\iota\phi'}\sin\frac{\theta'}{2}\,|1\rangle$ we have:

$$0 = \langle\psi|\psi'\rangle = \cos\frac{\theta'}{2}\cos\frac{\theta}{2} + e^{\iota(\phi'-\phi)}\sin\frac{\theta'}{2}\sin\frac{\theta}{2} \tag{1-10}$$

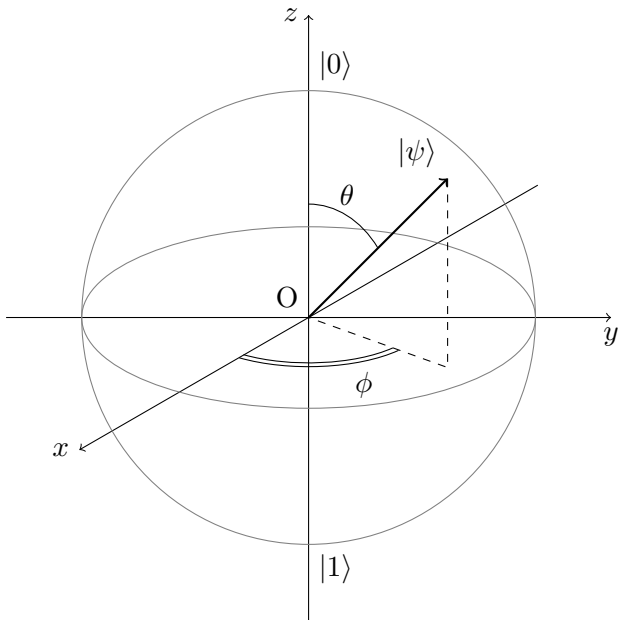

Figure 1: Representation in the Bloch sphere of the qubit state $|psi\rangle$.

For this equality to be satisfied, a necessary condition is that $e^{\iota(\phi'-\phi)}$ is real. This could happen for $\phi = \phi'$, in which case $e^{\iota(\phi'-\phi)} = 1$, then:

$$\cos\frac{\theta'}{2}\cos\frac{\theta}{2} + \sin\frac{\theta'}{2}\sin\frac{\theta}{2} = 0 \tag{1-11}$$

$$-\frac{1}{\tan\frac{\theta'}{2}} = \tan\frac{\theta}{2} \tag{1-12}$$

This equality is satisfied for $\frac{\theta'}{2} = \frac{\pi}{2} + \frac{\theta}{2}$, but in this way $\theta' > \pi$ and we are out of the domain assumed for this angle. Another solution is $\phi' = \phi + \pi$ so that $e^{\iota(\phi'-\phi)} = -1$; in this case:

$$\cos\frac{\theta'}{2}\cos\frac{\theta}{2} - \sin\frac{\theta'}{2}\sin\frac{\theta}{2} = 0 \tag{1-13}$$

$$\frac{1}{\tan\frac{\theta'}{2}} = \tan\frac{\theta}{2} \tag{1-14}$$

The above equation is satisfied for $\frac{\theta'}{2} = \frac{\pi}{2} - \frac{\theta}{2}$. This is a feasible solution, and corresponds in summary to:

$$\theta' = \pi - \theta \qquad\qquad \phi' = \phi + \pi$$

which is indeed the opposite vector in the Bloch sphere.

2 Observables quantities

- The quantum-mechanics formalism adopted for the qubit is perfectly overlapped to the formalism adopted for a spin-1/2 particle. Important observables in the qubit space are those associated to the **Pauli operators**. The Pauli operators are defined as:

$$\hat{\sigma}_x = |1\rangle\langle 0| + |0\rangle\langle 1| \tag{2-1}$$
$$\hat{\sigma}_y = \iota(|1\rangle\langle 0| - |0\rangle\langle 1|) \tag{2-2}$$
$$\hat{\sigma}_z = |0\rangle\langle 0| - |1\rangle\langle 1| \tag{2-3}$$

or in matrix form:

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \qquad \sigma_y = \begin{bmatrix} 0 & -\iota \\ \iota & 0 \end{bmatrix} \qquad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \tag{2-4}$$

Pauli operators are Hermitian:

$$\hat{\sigma}_x = \hat{\sigma}_x^\dagger \qquad \hat{\sigma}_y = \hat{\sigma}_y^\dagger \qquad \hat{\sigma}_z = \hat{\sigma}_z^\dagger \tag{2-5}$$

- The states of the computational basis are the **eigenstates** of $\hat{\sigma}_z$ operator, associated to the eigenvalues +1 and -1.

$$\hat{\sigma}_z |0\rangle = +1 \cdot |0\rangle \tag{2-6}$$
$$\hat{\sigma}_z |1\rangle = -1 \cdot |1\rangle \tag{2-7}$$

The **probabilities** to measure the states $|0\rangle$ or $|1\rangle$ are given by the squared modulus of their coefficients:

$$p(|0\rangle) = \cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2} \tag{2-8}$$
$$p(|1\rangle) = \sin^2\frac{\theta}{2} = \frac{1-\cos\theta}{2} \tag{2-9}$$

- Note that, despite the fact that the quantum state of the qubit may be a generic superposition of $|0\rangle$ and $|1\rangle$, the result of the quantum measurement is always *either* of the two eigenvalues, which in turn we associate to 0 or 1 as a logical value. Upon measurement, to each qubit corresponds one (and only one) bit of information.

- The expectation values of the quantities associated to the three Pauli operators are equal to the three cartesian coordinates of the vector in the Euclidean space associated to the qubit state:

$$\langle\hat{\sigma}_x\rangle = \langle\psi|\,\hat{\sigma}_x\,|\psi\rangle = x \tag{2-10}$$
$$\langle\hat{\sigma}_y\rangle = \langle\psi|\,\hat{\sigma}_y\,|\psi\rangle = y \tag{2-11}$$
$$\langle\hat{\sigma}_z\rangle = \langle\psi|\,\hat{\sigma}_z\,|\psi\rangle = z \tag{2-12}$$

Proof. We show the passages of the calculation for $\hat{\sigma}_z$. For the other two, one proceeds analogously.

$$\begin{aligned}\langle\hat{\sigma}_z\rangle = \langle\psi|\,\hat{\sigma}_z\,|\psi\rangle &= \left(\cos\frac{\theta}{2}\langle 0| + e^{-\iota\phi}\sin\frac{\theta}{2}\langle 1|\right)(|0\rangle\langle 0| - |1\rangle\langle 1|)\left(\cos\frac{\theta}{2}|0\rangle + e^{\iota\phi}\sin\frac{\theta}{2}|1\rangle\right) = \\ &= \left(\cos\frac{\theta}{2}\langle 0| - e^{-\iota\phi}\sin\frac{\theta}{2}\langle 1|\right)\left(\cos\frac{\theta}{2}|0\rangle + e^{\iota\phi}\sin\frac{\theta}{2}|1\rangle\right) = \\ &= \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = \cos\theta = z\end{aligned} \tag{2-13}$$

- The three Pauli operators together with the identity operator form a complete basis for the operators on the Hilbert space of the qubit. Namely, any operator in the qubit space can be written as a linear combination with complex coefficients of the three Pauli operators and of the identity operator.

  Proof. It is more convenient to use the operators written in matrix form. The matrices of the Pauli operators are written in Eq. (2-4). The matrix form of the identity operator is obviously $\mathbb{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. The proposition is proven by reshaping these matrices into column vectors, by assembling such vectors one next to the other to form another 4×4 square matrix:

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -\iota & 0 \\ 0 & 1 & \iota & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix} \tag{2-14}$$

  and by showing that the latter matrix has full rank. Namely, we need to show that the matrix (2-14) has non-zero determinant.

$$\det\left(\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -\iota & 0 \\ 0 & 1 & \iota & 0 \\ 1 & 0 & 0 & -1 \end{bmatrix}\right) = 1 \cdot \begin{vmatrix} 1 & -\iota & 0 \\ 1 & \iota & 0 \\ 0 & 0 & -1 \end{vmatrix} - 1 \cdot \begin{vmatrix} 0 & 1 & -\iota \\ 0 & 1 & \iota \\ 1 & 0 & 0 \end{vmatrix} =$$
$$= -1 \cdot \begin{vmatrix} 1 & -\iota \\ 1 & \iota \end{vmatrix} - -1 \cdot \begin{vmatrix} 1 & -\iota \\ 1 & \iota \end{vmatrix} = -2 \cdot \begin{vmatrix} 1 & -\iota \\ 1 & \iota \end{vmatrix} = -2 \cdot (\iota + \iota) = -4\iota \neq 0 \tag{2-15}$$

- Observables are associated to Hermitian operators. Any observable in the qubit space is written as a linear combination with *real coefficients* of the three Pauli operators and of the identity operator.

  Proof. Let $\hat{q}$ be the generic operator written as linear combination of the three Pauli operators and of the identity operator:

$$\hat{q} = \alpha\hat{\sigma}_x + \beta\hat{\sigma}_y + \gamma\hat{\sigma}_z + \delta\mathbb{I} \tag{2-16}$$

  For $\hat{q}$ to be Hermitian we need:

$$\hat{q}^\dagger = \hat{q} \tag{2-17}$$
$$\alpha^*\hat{\sigma}_x + \beta^*\hat{\sigma}_y + \gamma^*\hat{\sigma}_z + \delta^*\mathbb{I} = \alpha\hat{\sigma}_x + \beta\hat{\sigma}_y + \gamma\hat{\sigma}_z + \delta\mathbb{I} \tag{2-18}$$

  This implies:

$$\alpha^* = \alpha \qquad \beta^* = \beta \qquad \gamma^* = \gamma \qquad \delta^* = \delta \tag{2-19}$$

  Namely, the four coefficients must be real numbers.

- The most generic traceless operator in the qubit space is a linear combination of $\hat{\sigma}_x$, $\hat{\sigma}_y$ and $\hat{\sigma}_z$.

  Proof. The trace of the generic operator written as in Eq. (2-16) is:

$$\mathrm{Tr}\,(\hat{q}) = \mathrm{Tr}\,(\alpha\hat{\sigma}_x + \beta\hat{\sigma}_y + \gamma\hat{\sigma}_z + \delta\mathbb{I}) =$$
$$= \alpha\mathrm{Tr}(\hat{\sigma}_x) + \beta\mathrm{Tr}(\hat{\sigma}_y) + \gamma\mathrm{Tr}(\hat{\sigma}_z) + \delta\mathrm{Tr}(\mathbb{I}) = \hat{\alpha} \cdot 0 + \beta \cdot 0 + \gamma \cdot 0 + \delta \cdot 2 = 2\delta \tag{2-20}$$

  Hence, in order to have $\mathrm{Tr}\,(\hat{q}) = 0$ we must have $\delta = 0$.

- Given the above propositions, the most generic traceless Hermitian operator takes the form:

$$\hat{q} = \vec{v} \cdot \hat{\vec{\sigma}} = \begin{bmatrix} v_x & v_y & v_z \end{bmatrix} \cdot \begin{bmatrix} \hat{\sigma}_x \\ \hat{\sigma}_y \\ \hat{\sigma}_z \end{bmatrix} \tag{2-21}$$

  where $\hat{v} \in \mathbb{R}^3$.

Its expectation value is:

$$\langle \hat{q} \rangle = \langle \vec{v} \cdot \hat{\vec{\sigma}} \rangle = \vec{v} \cdot \langle \hat{\vec{\sigma}} \rangle = v_x x + v_y y + v_z z \tag{2-22}$$

where $x = \langle \hat{\sigma}_x \rangle$, $y = \langle \hat{\sigma}_y \rangle$ and $z = \langle \hat{\sigma}_z \rangle$.

- The operator:

$$\hat{\sigma}_{\vec{u}} = \vec{u} \cdot \hat{\vec{\sigma}} \tag{2-23}$$

with $\vec{u} \in \mathbb{R}^3$ and $|\vec{u}| = 1$ generalizes the concept of Pauli operator, in relation to an arbitrary axis directed as the unit vector $\vec{u}$. The physical meaning of the observable associated to $\hat{\sigma}_{\vec{u}}$ is readily noted, by observing from Eq. (2-22) that the expectation value of this operator is the projection, on the direction of $\vec{u}$, of the vector associated to the qubit state, which is $\langle \hat{\vec{\sigma}} \rangle = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

If we write:

$$\vec{u} = \begin{bmatrix} \cos\phi \sin\theta \\ \sin\phi \sin\theta \\ \cos\theta \end{bmatrix} \tag{2-24}$$

the $\hat{\sigma}_{\vec{u}}$ operator in matrix form reads:

$$\sigma_{\vec{u}} = \begin{bmatrix} \cos\theta & e^{-\iota\phi} \sin\theta \\ e^{+\iota\phi} \sin\theta & -\cos\theta \end{bmatrix} \tag{2-25}$$

- Eigenstates of $\hat{\sigma}_{\vec{u}}$ are:

$$|0_{\vec{u}}\rangle = \cos\frac{\theta}{2} |0\rangle + e^{+\iota\phi} \sin\frac{\theta}{2} |1\rangle \tag{2-26}$$

$$|1_{\vec{u}}\rangle = \sin\frac{\theta}{2} |0\rangle - e^{+\iota\phi} \cos\frac{\theta}{2} |1\rangle \tag{2-27}$$

associated respectively to the eigenvalues -1 and $+1$. Note that $|0_{\vec{u}}\rangle = |1_{-\vec{u}}\rangle$.

Proof. Eigenvalues and eigenvectors could be retrieved with standard algebraic procedures from the matrix form Eq. (2-25). Here we adopt another procedure. We observe that, as any Hermitian operator in the qubit space, which is two-dimensional, also $\hat{\sigma}_{\vec{u}}$ needs to have two eigenvalues λ_0 and λ_1 and two eigenstates $|0_{\vec{u}}\rangle$ and $|1_{\vec{u}}\rangle$. Writing a generic qubit state in this basis as $|\psi\rangle = \alpha |0_{\vec{u}}\rangle + \beta |1_{\vec{u}}\rangle$, the expectation value $\langle \hat{\sigma}_{\vec{u}} \rangle$ is:

$$\langle \hat{\sigma}_{\vec{u}} \rangle = |\alpha|^2 \lambda_0 + |\beta|^2 \lambda_1 \tag{2-28}$$

Taking $\lambda_0 > \lambda_1$, the maximum value $\langle \hat{\sigma}_{\vec{u}} \rangle = \lambda_0$ is reached for $|\psi\rangle = |0_{\vec{u}}\rangle$ while the minimum value $\langle \hat{\sigma}_{\vec{u}} \rangle = \lambda_1$ is reached for $|\psi\rangle = |1_{\vec{u}}\rangle$.

From Eq. (2-22) we see that the expectation value for this kind of operators equals a scalar product between two Euclidean vectors. As well known, the scalar product reaches the maximum and minimum values respectively for parallel and antiparallel vectors.

In this case, as both are unit vectors, maximum and minimum values are +1 and -1. Therefore, these are the eigenvalues of $\hat{\sigma}_{\vec{u}}$. The related eigenvectors are those qubit states associated to the vector $\vec{u}$ and to the vector $-\vec{u}$ respectively, which are indeed the kets written above as $|0_{\vec{u}}\rangle$ and $|1_{\vec{u}}\rangle$.

- Given the expression of the eigenvalues and eigenvectors, the operator $\hat{\sigma}_{\vec{u}}$ is written in diagonal form as:

$$\hat{\sigma}_{\vec{u}} = |0_{\vec{u}}\rangle \langle 0_{\vec{u}}| - |1_{\vec{u}}\rangle \langle 1_{\vec{u}}| \tag{2-29}$$

3 Measuring qubits and the no-cloning theorem

- **The quantum state of a single qubit in an unknown state cannot be determined with certainty by means of experimental measurements.** In other words, the coefficients α and β determining its state cannot be characterized with arbitrary precision. In fact, any measurement in the qubit space gives as outcome one of the two eigenvalues of the associated operator, which is only 1 bit of information. The probability to obtain one or the other eigenvalue depends of course on α and β. However, this measurement does not correspond to their value. Note that after the measurement the qubit state is collapsed on the related eigenstates, and repeating the measurement would again produce the same eigenvalue, without adding information on the initial state.

- Instead, the coefficients α and β may be estimated with arbitrary precision if we have **an arbitrarily large number of qubits identically prepared** in the same state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. Say that we have N identical qubits and we measure $\hat{\sigma}_z$ for each of them. We shall obtain N_0 times the eigenvalue $+1$ (related to the logic 0) and N_1 times the eigenvalue -1 (related to the logic 1), with $N_0 + N_1 = N$. In the limit of large N, one has:

$$\frac{N_0}{N} \simeq p(|0\rangle) = |\alpha|^2 \qquad \frac{N_1}{N} \simeq p(|1\rangle) = |\beta|^2 \tag{3-1}$$

  Therefore, in the same limit one can estimate $\langle\hat{\sigma}_z\rangle = z$ as:

$$\frac{N_0}{N} - \frac{N_1}{N} \simeq |\alpha|^2 - |\beta|^2 = \langle\hat{\sigma}_z\rangle = z \tag{3-2}$$

  Having other two sets of identically prepared qubits (for a total of $3N$ identical qubits) we could measure $\hat{\sigma}_x$ and $\hat{\sigma}_y$ and produce analogous estimates of x and y. The knowledge of the three components x, y and z corresponds to the knowledge of the vector in the Bloch sphere, and thus to the knowledge of the full qubit state with its coefficients.

- If a method existed for making a prefect copy (a 'clone') of qubit in an unknown state, the limitation discussed above could be circumvented. Indeed, from a single qubit we could make a lot of identical copies (say $3N$) and we could proceed with the measurements of $\hat{\sigma}_x$, $\hat{\sigma}_y$ and $\hat{\sigma}_z$, finally retrieving the qubit state. As a matter of fact, quantum mechanics intrinsically prevents this possibility, as stated by the following 'no-cloning' theorem.

- **No-cloning theorem** It is impossible to build a machine that operates unitary transformations and is able to clone the generic state of a qubit.

  Proof.

  - Let us suppose that such a machine exists, working on two qubits A and B. It would be able to transform $|\phi\rangle_A|0\rangle_B$ into $|\phi\rangle_A|\phi\rangle_B$, and $|\psi\rangle_A|0\rangle_B$ into $|\psi\rangle_A|\psi\rangle_B$, for any pair of generic states $|\phi\rangle$ and $|\psi\rangle$, and by performing the same unitary transformation $\hat{U}$. In formulas:

$$\hat{U}|\phi\rangle_A|0\rangle_B = |\phi\rangle_A|\phi\rangle_B \tag{3-3}$$
$$\hat{U}|\psi\rangle_A|0\rangle_B = |\psi\rangle_A|\psi\rangle_B \tag{3-4}$$

  - We take the Hermitian conjugate of the first relation:

$$\langle\phi|_A\langle 0|_B\hat{U}^\dagger = \langle\phi|_A\langle\phi|_B$$

    and we evaluate the bra-ket product with the second one:

$$\langle\phi|_A\langle 0|_B\hat{U}^\dagger\hat{U}|\psi\rangle_A|0\rangle_B = \langle\phi|_A\langle\phi|_B|\psi\rangle_A|\psi\rangle_B \tag{3-5}$$

Considering that $\hat{U}$ is unitary and thus $\hat{U}^\dagger \hat{U} = \mathbb{I}$:

$$\langle\phi|\psi\rangle\langle 0|0\rangle = \langle\phi|\psi\rangle\langle\phi|\psi\rangle \tag{3-6}$$

$$\langle\phi|\psi\rangle = (\langle\phi|\psi\rangle)^2 \tag{3-7}$$

$$\langle\phi|\psi\rangle\,(1 - \langle\phi|\psi\rangle) = 0 \tag{3-8}$$

which has two solutions:

$$\langle\phi|\psi\rangle = 1 \qquad\qquad \langle\phi|\psi\rangle = 0 \tag{3-9}$$

– This means that a unitary $\hat{U}$ could clone qubit states $|\phi\rangle$ and $|\psi\rangle$ only in two cases. First, if the two states are actually the same ($\langle\phi|\psi\rangle = 1$), namely the unitary is able to clone only one specific state. Second, if the two states are orthogonal ($\langle\phi|\psi\rangle = 0$), namely the unitary is able to clone two specific orthogonal states. In conclusion, it is not possible to conceive a transformation $\hat{U}$ that clones generic states of the qubit, but only at most two specific orthogonal states.

- In a more complex scenario, one may consider a unitary $\hat{U}$ that not only acts on the two qubits (the one to be cloned and the target one), but also on other arbitrary physical systems (the environment, or the machine itself performing the transformation). As a matter of fact, this further hypothesis does not alter the result of the no-cloning theorem, as it is easily shown. Say that the environment is initially in a state $|\mathcal{A}\rangle_E$; we proceed as in the proof of the theorem and write the operation of $\hat{U}$ as:

$$\hat{U}\,|\phi\rangle_A\,|0\rangle_B\,|\mathcal{A}\rangle_E = |\phi\rangle_A\,|\phi\rangle_B\,|\mathcal{A}_\phi\rangle_E \tag{3-10}$$

$$\hat{U}\,|\psi\rangle_A\,|0\rangle_B\,|\mathcal{A}\rangle_E = |\psi\rangle_A\,|\psi\rangle_B\,|\mathcal{A}_\psi\rangle_E \tag{3-11}$$

Note that we are allowing the environment state to change in a different way, depending on the state of the qubit to be cloned. From these relations we retrieve:

$$\langle\phi|\psi\rangle\langle 0|0\rangle\langle\mathcal{A}|\mathcal{A}\rangle = \langle\phi|\psi\rangle\langle\phi|\psi\rangle\langle\mathcal{A}_\phi|\mathcal{A}_\psi\rangle \tag{3-12}$$

$$\langle\phi|\psi\rangle\,(1 - \langle\phi|\psi\rangle\langle\mathcal{A}_\phi|\mathcal{A}_\psi\rangle) = 0 \tag{3-13}$$

Again, one possible solution is $\langle\phi|\psi\rangle = 0$ which has the same interpretation as before: the machine can clone two specific orthogonal states. Another solution is $\langle\phi|\psi\rangle\langle\mathcal{A}_\phi|\mathcal{A}_\psi\rangle = 1$. The bra-ket product between two normalized kets, in general, is a complex number with modulus smaller or equal to one. For their product to be equal to one, both moduli should be equal to one, with the possible freedom of a phase term:

$$|\psi\rangle = e^{\iota\theta}\,|\phi\rangle \qquad\qquad |\mathcal{A}_\psi\rangle = e^{-\iota\theta}\,|\mathcal{A}_\phi\rangle \tag{3-14}$$

This second solution amounts to $|\psi\rangle$ and $|\phi\rangle$ being the same state, apart from an irrelevant phase term. Therefore, there is still no possibility of cloning arbitrary and unknown qubit states.

4 Single-qubit gates: general definition and notable examples

- In the classical-information framework, only two operations can be conceived that act on a single bit:
  - The identity gate, which leaves the bit value unaltered.

$$\begin{array}{ccc} 0 & \rightarrow & 0 \\ 1 & \rightarrow & 1 \end{array}$$

  - The NOT gate, which flips the bit value.

$$\begin{array}{ccc} 0 & \rightarrow & 1 \\ 1 & \rightarrow & 0 \end{array}$$

- In the framework of quantum information, one instead may conceive an infinity of operators that can be applied to the state of the two-level quantum system. The only constraint is that they need to **keep proper normalization of the state**.

  Let $|\psi'\rangle = \hat{U}\,|\psi\rangle$; then, in order to preserve the norm:

$$\langle\psi'|\psi'\rangle = \langle\psi|\,\hat{U}^\dagger\hat{U}\,|\psi\rangle = \langle\psi|\psi\rangle \tag{4-1}$$

  so it must be $\hat{U}^\dagger\hat{U} = \mathbb{I}$. An operator $\hat{U}$ that satisfies this condition is by definition a **unitary operator**.

Single-qubit gates
Single-qubit gates are all the unitary operators acting on the Hilbert space of a single qubit.

- One trivial single-qubit gate is the **identity operator** $\mathbb{I}$.

- Notable single-qubit gates are the **Pauli operators** $\hat{\sigma}_x$, $\hat{\sigma}_y$ and $\hat{\sigma}_y$. Indeed, besides being Hermitian these operators are also unitary. The Pauli operators have been already partially discussed as observables in the qubit space; interesting properties are related to their products, commutators and anticommutators.
  - As they are both Hermitian ($\hat{U} = \hat{U}^\dagger$) and unitary ($\hat{U}^\dagger = \hat{U}^{-1}$) they coincide with their inverse.

$$\hat{\sigma}_x^2 = \hat{\sigma}_y^2 = \hat{\sigma}_z^2 = \mathbb{I} \tag{4-2}$$

  - Their products are easily calculated from their matrix form and give:

$$\hat{\sigma}_x\hat{\sigma}_y = \iota\hat{\sigma}_z \qquad \hat{\sigma}_y\hat{\sigma}_z = \iota\hat{\sigma}_x \qquad \hat{\sigma}_z\hat{\sigma}_x = \iota\hat{\sigma}_y \tag{4-3}$$

    It follows immediately:[1]

$$\hat{\sigma}_y\hat{\sigma}_x = -\iota\hat{\sigma}_z \qquad \hat{\sigma}_z\hat{\sigma}_y = -\iota\hat{\sigma}_x \qquad \hat{\sigma}_x\hat{\sigma}_z = -\iota\hat{\sigma}_y \tag{4-4}$$

[1] It is sufficient to consider the Hermitian conjugate of the product, and to exploit the fact that the Pauli operators are Hermitian. For instance, the first relation is proven as follows:

$$\hat{\sigma}_y\hat{\sigma}_x = (\hat{\sigma}_x^\dagger\hat{\sigma}_y^\dagger)^\dagger = (\hat{\sigma}_x\hat{\sigma}_y)^\dagger = (\iota\hat{\sigma}_z)^\dagger = -\iota\hat{\sigma}_z^\dagger = -\iota\hat{\sigma}_z$$

The other two are proven in the same fashion.

- The commutators and anticommutators are calculated straightforwardly from the former relations on the products:

$$[\hat{\sigma}_x, \hat{\sigma}_y] = 2\iota\hat{\sigma}_z \qquad \{\hat{\sigma}_x, \hat{\sigma}_y\} = 0 \tag{4-5}$$
$$[\hat{\sigma}_y, \hat{\sigma}_z] = 2\iota\hat{\sigma}_x \qquad \{\hat{\sigma}_y, \hat{\sigma}_z\} = 0 \tag{4-6}$$
$$[\hat{\sigma}_z, \hat{\sigma}_x] = 2\iota\hat{\sigma}_y \qquad \{\hat{\sigma}_z, \hat{\sigma}_x\} = 0 \tag{4-7}$$

- The Pauli operators are generalized to an arbitrary direction by the operator $\hat{\sigma}_{\vec{u}} = \vec{u} \cdot \hat{\vec{\sigma}}$. This is also a Hermitian and unitary operator:

$$\hat{\sigma}_{\vec{u}}^2 = \mathbb{I} \tag{4-8}$$

- Another notable single-qubit gate, employed in many quantum algorithms, is the **Hadamard gate**:

$$\hat{U}_H = \frac{\hat{\sigma}_x + \hat{\sigma}_z}{\sqrt{2}} \equiv \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \tag{4-9}$$

Using bra-ket notation, it is written as:

$$\begin{aligned} \hat{U}_H &= \frac{1}{\sqrt{2}} \Big(|0\rangle \langle 0| + |0\rangle \langle 1| + |1\rangle \langle 0| - |1\rangle \langle 1| \Big) = \\ &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \langle 0| + \frac{|0\rangle - |1\rangle}{\sqrt{2}} \langle 1| \end{aligned} \tag{4-10}$$

This expression highlights the operation of the Hadamard gate on the states of the computational basis:

$$\hat{U}_H |0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \qquad \hat{U}_H |1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \tag{4-11}$$

The Hadamard gate is also both Hermitian and unitary, and thus coincides with its inverse:

$$\hat{U}_H^2 = \mathbb{I} \tag{4-12}$$

Hence:

$$\hat{U}_H \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |0\rangle \qquad \hat{U}_H \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |1\rangle \tag{4-13}$$

- Finally, it is worth mentioning the **phase gate**.

$$\hat{U}_{z,\delta} = \cos\frac{\delta}{2}\mathbb{I} - \iota \sin\frac{\delta}{2}\hat{\sigma}_z \equiv \begin{bmatrix} e^{-\iota\frac{\delta}{2}} & 0 \\ 0 & e^{+\iota\frac{\delta}{2}} \end{bmatrix} \tag{4-14}$$

As global phase terms are not relevant, another equivalent form for the same phase gate is:

$$\hat{U}_{z,\delta} \equiv \begin{bmatrix} 1 & 0 \\ 0 & e^{+\iota\delta} \end{bmatrix} \tag{4-15}$$

which is simply written in bra-ket notation as:

$$\hat{U}_{z,\delta} = |0\rangle \langle 0| + e^{\iota\delta} |1\rangle \langle 1| \tag{4-16}$$

Depending on the convenience, one or the other expression may be employed.

The phase gate amounts to a *rotation of the qubit* of an angle δ about the z axis, in the Bloch sphere. This is easily observed by considering the phase gate in the form of Eq. (4-16), acting on the generic qubit state Eq. (1-8):

$$\hat{U}_{z,\delta} \left(\cos\frac{\theta}{2} |0\rangle + e^{\iota\phi} \sin\frac{\theta}{2} |1\rangle \right) = \cos\frac{\theta}{2} |0\rangle + e^{\iota(\phi+\delta)} \sin\frac{\theta}{2} |1\rangle \tag{4-17}$$

It is clear that the only change in the qubit state is the azimuthal angle, which has increased from ϕ to $\phi + \delta$.

5 Representations and decompositions of single-qubit operations

- Before discussing in detail the unitary operators describing single-qubit gates, it is useful to recall a few properties of generic unitary operators. First, every unitary operator has *complex eigenvalues with unit modulus.*[2] Second, *unitary operators can always be diagonalized.* This allows to write a generic unitary operator $\hat{U}$ in the diagonal form as:

$$\hat{U} = \sum_i \lambda_i \left|e_i\right\rangle \left\langle e_i\right| \tag{5-2}$$

  where $|e_i\rangle$ are the eigenvectors, forming an orthonormal complete basis of the Hilbert space, and $\lambda_i = e^{\iota\varphi_i}$ are the related eigenvalues ($\varphi_i \in \mathbb{R}$). Third, unitary operators *transforms orthogonal vectors into orthogonal vectors.*[3]

- Single-qubit gates are unitary operators acting on a two-dimensional Hilbert space. A complete orthonormal basis of vectors of this Hilbert space can always be written as $|0_{\vec{u}}\rangle$ and $|1_{\vec{u}}\rangle$ for a suitable direction $\vec{u}$ in the Bloch sphere. Hence, every single-qubit gate can be written in **diagonal form** as:

$$\hat{U} = e^{\iota\varphi_0} \left|0_{\vec{u}}\right\rangle \left\langle 0_{\vec{u}}\right| + e^{\iota\varphi_1} \left|1_{\vec{u}}\right\rangle \left\langle 1_{\vec{u}}\right| \tag{5-4}$$

  As overall phase factors are not relevant, we may define $\delta = \varphi_1 - \varphi_0$ and write an equivalent expression for the same generic single-qubit gate as:

$$\hat{U}_{\vec{u},\delta} = \left|0_{\vec{u}}\right\rangle \left\langle 0_{\vec{u}}\right| + e^{\iota\delta} \left|1_{\vec{u}}\right\rangle \left\langle 1_{\vec{u}}\right| \tag{5-5}$$

  or also as:

$$\hat{U}_{\vec{u},\delta} = e^{-\iota\frac{\delta}{2}} \left|0_{\vec{u}}\right\rangle \left\langle 0_{\vec{u}}\right| + e^{+\iota\frac{\delta}{2}} \left|1_{\vec{u}}\right\rangle \left\langle 1_{\vec{u}}\right| \tag{5-6}$$

  By comparing Eq. (5-5) with Eq. (4-16) we understand that **every single-qubit gate is equivalent to a rotation of the qubit in the Bloch sphere** around a suitable axis $\vec{u}$, by an angle δ defined as above.

- In this perspective, the Pauli operator $\hat{\sigma}_z$ amounts to a π rotation around the z axis, as it reads:

$$\hat{\sigma}_z = \left|0\right\rangle \left\langle 0\right| - \left|1\right\rangle \left\langle 1\right| = \left|0\right\rangle \left\langle 0\right| + e^{\iota\pi} \left|1\right\rangle \left\langle 1\right| = \hat{U}_{z,\pi} \tag{5-7}$$

  In other words, $\hat{\sigma}_z$ is a phase gate with $\delta = \pi$.

  The other Pauli operators $\hat{\sigma}_x$ and $\hat{\sigma}_y$ are π rotations around respectively the x and y axis; the $\hat{\sigma}_{\vec{u}}$ operator is a π rotation around the generic $\vec{u}$ axis:

$$\hat{\sigma}_{\vec{u}} = \left|0_{\vec{u}}\right\rangle \left\langle 0_{\vec{u}}\right| - \left|1_{\vec{u}}\right\rangle \left\langle 1_{\vec{u}}\right| = \hat{U}_{\vec{u},\pi} \tag{5-8}$$

- Exploiting Eq. (5-6) and the Euler's formula for the exponentials, we can derive another way of writing the generic gate. We start by the decomposing the coefficients $e^{\pm\iota\delta/2}$ in their real and imaginary parts.

$$\hat{U}_{\vec{u},\delta} = \cos\left(-\frac{\delta}{2}\right) \left|0_{\vec{u}}\right\rangle \left\langle 0_{\vec{u}}\right| + \iota \sin\left(-\frac{\delta}{2}\right) \left|0_{\vec{u}}\right\rangle \left\langle 0_{\vec{u}}\right| + \cos\left(\frac{\delta}{2}\right) \left|1_{\vec{u}}\right\rangle \left\langle 1_{\vec{u}}\right| + \iota \sin\left(\frac{\delta}{2}\right) \left|1_{\vec{u}}\right\rangle \left\langle 1_{\vec{u}}\right| \tag{5-9}$$

[2] For a given eigenvalue λ and related eigenvector $|e\rangle$, one has $\hat{U}\left|e\right\rangle = \lambda\left|e\right\rangle$. The Hermitian conjugate of this relation gives $\left\langle e\right|\hat{U}^\dagger = \left\langle e\right|\lambda^*$. The normalization condition on the ket $|\psi'\rangle = \hat{U}\left|e\right\rangle$ thus implies:

$$1 = \langle\psi'|\psi'\rangle = \left\langle e\right|\hat{U}^\dagger\hat{U}\left|e\right\rangle = \left\langle e\right|\lambda^*\lambda\left|e\right\rangle = |\lambda|^2\langle e|e\rangle = |\lambda|^2 \tag{5-1}$$

[3] Let $|\psi\rangle$ and $|\phi\rangle$ be orthogonal kets, i.e. $\langle\psi|\phi\rangle = 0$. Let the transformed kets be $|\psi'\rangle = \hat{U}\left|\psi\right\rangle$ and $|\phi'\rangle = \hat{U}\left|\phi\right\rangle$. Then,

$$\langle\psi'|\phi'\rangle = \left\langle\psi\right|\hat{U}^\dagger\hat{U}\left|\phi\right\rangle = \left\langle\psi\right|\mathbb{I}\left|\phi\right\rangle = \langle\psi|\phi\rangle = 0 \tag{5-3}$$

which means that the transformed kets $|\psi'\rangle$ and $|\phi'\rangle$ are also orthogonal.

By recalling that $\cos\left(-\frac{\delta}{2}\right) = \cos\left(+\frac{\delta}{2}\right)$ and $\sin\left(-\frac{\delta}{2}\right) = -\sin\left(+\frac{\delta}{2}\right)$, we rewrite:

$$\hat{U}_{\vec{u},\delta} = \cos\left(\frac{\delta}{2}\right)\left[|0_{\vec{u}}\rangle\langle 0_{\vec{u}}| + |1_{\vec{u}}\rangle\langle 1_{\vec{u}}|\right] - \iota\sin\left(\frac{\delta}{2}\right)\left[|0_{\vec{u}}\rangle\langle 0_{\vec{u}}| - |1_{\vec{u}}\rangle\langle 1_{\vec{u}}|\right] \tag{5-10}$$

The identity operator in the basis of $|0_{\vec{u}}\rangle$ and $|1_{\vec{u}}\rangle$ is simply $\mathbb{I} = |0_{\vec{u}}\rangle\langle 0_{\vec{u}}| + |1_{\vec{u}}\rangle\langle 1_{\vec{u}}|$. Therefore, the generic single-qubit gate can also be written as:

$$\hat{U}_{\vec{u},\delta} = \cos\left(\frac{\delta}{2}\right)\mathbb{I} - \iota\sin\left(\frac{\delta}{2}\right)\hat{\sigma}_{\vec{u}} \tag{5-11}$$

- Every single-qubit gate can be written in the **exponential form**.

$$\hat{U}_{\vec{u},\delta} = e^{-\iota\frac{\delta}{2}\hat{\sigma}_{\vec{u}}} \tag{5-12}$$

Proof. We write the operator $\hat{U}_{\vec{u},\delta}$ in matrix form, choosing the basis of the vectors $|0_{\vec{u}}\rangle$ and $|1_{\vec{u}}\rangle$; from Eq. (5-6) we have:

$$\hat{U}_{\vec{u},\delta} \equiv \begin{bmatrix} e^{-\iota\frac{\delta}{2}} & 0 \\ 0 & e^{+\iota\frac{\delta}{2}} \end{bmatrix}_{\vec{u}} \tag{5-13}$$

The subscript $\vec{u}$ to the matrix is just to recall that the matrix is written in that basis. Due to the properties of diagonal matrices, we can write:

$$\begin{bmatrix} e^{-\iota\frac{\delta}{2}} & 0 \\ 0 & e^{+\iota\frac{\delta}{2}} \end{bmatrix}_{\vec{u}} = \exp\left(\begin{bmatrix} -\iota\frac{\delta}{2} & 0 \\ 0 & +\iota\frac{\delta}{2} \end{bmatrix}_{\vec{u}}\right) = \exp\left(-\iota\frac{\delta}{2}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}_{\vec{u}}\right) \tag{5-14}$$

But precisely we have $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}_{\vec{u}} \equiv \hat{\sigma}_{\vec{u}}$, which proves the proposition.

The exponential form can be useful in order to associate a single-qubit gate to a physical Hamiltonian. Indeed, the evolution of a quantum system under the Hamiltonian $\hat{H}$ proceeds according to the time-evolution operator $\hat{U} = e^{-\iota\hat{H}t/\hbar}$.

- As single-qubit gates correspond to three-dimensional rotations in the Bloch sphere, they follow analogous rules when applied in sequence. Rotations about the same axis can be composed directly and are commutative.

$$\hat{U}_{\vec{u},\alpha}\hat{U}_{\vec{u},\beta} = \hat{U}_{\vec{u},\beta}\hat{U}_{\vec{u},\alpha} = e^{-\iota(\alpha+\beta)\hat{\sigma}_{\vec{u}}} = \hat{U}_{\vec{u},\alpha+\beta} \tag{5-15}$$

However, generic rotations about different axes are not commutative.

$$\hat{U}_{\vec{u},\alpha}\hat{U}_{\vec{v},\beta} \neq \hat{U}_{\vec{v},\beta}\hat{U}_{\vec{u},\alpha} \tag{5-16}$$

- There are several ways of decomposing a given gate into a product, i.e. a sequence, of gates of different kind. In this respect, a very significant result is the following. **Every single-qubit gate can be decomposed into a sequence of three qubit rotations respectively about z, y and again z.** Accordingly, it can be univocally identified by the three angles of these rotations.

$$\hat{U} = \hat{U}_{z,\alpha}\hat{U}_{y,\beta}\hat{U}_{z,\gamma} = \hat{U}_{\alpha,\beta,\gamma} \tag{5-17}$$

Proof.

  - Single-qubit gates are unitary transformations and transform orthogonal qubit states into orthogonal qubit states. The transformation is fully defined by how it transforms a complete set of orthogonal states.

- Two orthogonal qubit states, composing a complete basis of the Hilbert space of the qubits, are opposite vectors in the Bloch sphere. Note that they may be both defined by means of a single euclidean vector $\vec{u}$ (the two orthogonal states are the one parallel and the one antiparallel to $\vec{u}$). An arbitrary single-qubit gate corresponds to a three-dimensional rotation that brings a vector $\vec{u}$ into a vector $\vec{v}$, and may be unequivocally defined by this condition.
- In principle, in the Euclidean three-dimensional space, there would be infinitely many different rotations that satisfy such a condition. Indeed, one vector is clearly not sufficient to identify a basis of the three-dimensional space. However, in the case of the qubits the vector $\vec{u}$ is sufficient to identify the basis. Thus, each of these rotations must be an equivalent representation of the single-qubit gate it is associated to. We may focus on only one of them.
- We can write the kets associated to a vector $\vec{u}$ and a vector $\vec{v}$ respectively as:

$$|u\rangle = \cos\frac{\theta_1}{2}|0\rangle + e^{\iota\phi_1}\sin\frac{\theta_1}{2}|1\rangle \tag{5-18}$$

$$|v\rangle = \cos\frac{\theta_2}{2}|0\rangle + e^{\iota\phi_2}\sin\frac{\theta_2}{2}|1\rangle \tag{5-19}$$

To bring the vector $|u\rangle$ in $|v\rangle$ we can act with the following sequence of three rotations in the Bloch sphere:

1. Apply $\hat{U}_{z,-\phi_1}$ which brings $\vec{u}$ in the xOz plane:

$$\hat{U}_{z,-\phi_1}|u\rangle = \cos\frac{\theta_1}{2}|0\rangle + \sin\frac{\theta_1}{2}|1\rangle \tag{5-20}$$

2. Apply $\hat{U}_{y,\theta_2-\theta_1}$ which adjusts the θ angle:

$$\hat{U}_{y,\theta_2-\theta_1}\hat{U}_{z,-\phi_1}|u\rangle = \cos\frac{\theta_2}{2}|0\rangle + \sin\frac{\theta_2}{2}|1\rangle \tag{5-21}$$

3. Apply $\hat{U}_{z,\phi_2}$ which finally sets the ϕ_2 angle and brings the qubit state to $|v\rangle$:

$$\hat{U}_{z,\phi_2}\hat{U}_{y,\theta_2-\theta_1}\hat{U}_{z,-\phi_1}|u\rangle = \cos\frac{\theta_2}{2}|0\rangle + e^{\iota\phi_2}\sin\frac{\theta_2}{2}|1\rangle = |v\rangle \tag{5-22}$$

This proves that an arbitrary qubit rotation takes the form $\hat{U}_{z,\alpha}\hat{U}_{y,\beta}\hat{U}_{z,\gamma} = \hat{U}_{\alpha,\beta,\gamma}$. Here, with reference to $\vec{u}$ and $\vec{v}$, the angles are $\alpha = \phi_2$, $\beta = \theta_2 - \theta_1$, and $\gamma = -\phi_1$.

This results means that a quantum computer is able to produce all possible single-qubit transformations if it is able to produce all qubit rotations about two fixed orthogonal axes in the Bloch sphere (which we may arbitrarily label y and z). Note that the three rotations in the decomposition do not commute.

- **Rotations in the Bloch sphere about the y axis can be decomposed into a sequence of Hadamard gates and phase gates (i.e. rotations about the z axis).**

$$\hat{U}_{y,\beta} = \hat{U}_{z,\frac{\pi}{2}}\,\hat{U}_H\,\hat{U}_{z,\beta}\,\hat{U}_H\,\hat{U}_{z,-\frac{\pi}{2}} \tag{5-23}$$

Proof.

- Let us write $\hat{U}_H\,\hat{U}_{z,\beta}\,\hat{U}_H$ in matrix form and let us elaborate the product:

$$\begin{aligned}
\hat{U}_H\,\hat{U}_{z,\beta}\,\hat{U}_H &\equiv \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\cdot\begin{bmatrix}e^{-\iota\frac{\beta}{2}} & 0\\ 0 & e^{+\iota\frac{\beta}{2}}\end{bmatrix}\cdot\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix} = \frac{1}{2}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\cdot\begin{bmatrix}e^{-\iota\frac{\beta}{2}} & e^{-\iota\frac{\beta}{2}}\\ e^{+\iota\frac{\beta}{2}} & e^{+\iota\frac{\beta}{2}}\end{bmatrix} = \\
&= \frac{1}{2}\begin{bmatrix}e^{-\iota\frac{\beta}{2}} + e^{\iota\frac{\beta}{2}} & e^{-\iota\frac{\beta}{2}} - e^{\iota\frac{\beta}{2}}\\ e^{+\iota\frac{\beta}{2}} - e^{-\iota\frac{\beta}{2}} & e^{+\iota\frac{\beta}{2}} + e^{\iota\frac{\beta}{2}}\end{bmatrix} = \begin{bmatrix}\cos\frac{\beta}{2} & -\iota\sin\frac{\beta}{2}\\ -\iota\sin\frac{\beta}{2} & \cos\frac{\beta}{2}\end{bmatrix} = \\
&= \begin{bmatrix}1 & 0\\ 0 & -\iota\end{bmatrix}\begin{bmatrix}\cos\frac{\beta}{2} & -\sin\frac{\beta}{2}\\ \sin\frac{\beta}{2} & \cos\frac{\beta}{2}\end{bmatrix}\begin{bmatrix}1 & 0\\ 0 & \iota\end{bmatrix}
\end{aligned} \tag{5-24}$$

– We may then recall that:

$$\hat{U}_{z,-\frac{\pi}{2}} \equiv \begin{bmatrix} e^{\iota\frac{\pi}{4}} & 0 \\ 0 & e^{-\iota\frac{\pi}{4}} \end{bmatrix} = e^{\iota\frac{\pi}{4}} \begin{bmatrix} 1 & 0 \\ 0 & e^{-\iota\frac{\pi}{2}} \end{bmatrix} = e^{\iota\frac{\pi}{4}} \begin{bmatrix} 1 & 0 \\ 0 & -\iota \end{bmatrix} \tag{5-25}$$

$$\hat{U}_{z,+\frac{\pi}{2}} \equiv \begin{bmatrix} e^{-\iota\frac{\pi}{4}} & 0 \\ 0 & e^{\iota\frac{\pi}{4}} \end{bmatrix} = e^{-\iota\frac{\pi}{4}} \begin{bmatrix} 1 & 0 \\ 0 & e^{\iota\frac{\pi}{2}} \end{bmatrix} = e^{-\iota\frac{\pi}{4}} \begin{bmatrix} 1 & 0 \\ 0 & \iota \end{bmatrix} \tag{5-26}$$

$$\hat{U}_{y,\beta} = \cos\frac{\beta}{2}\mathbb{I} - \iota\sin\frac{\beta}{2}\hat{\sigma}_y \equiv \begin{bmatrix} \cos\frac{\beta}{2} & -\sin\frac{\beta}{2} \\ \sin\frac{\beta}{2} & \cos\frac{\beta}{2} \end{bmatrix} \tag{5-27}$$

Therefore, the previous expression reads:

$$\hat{U}_H\,\hat{U}_{z,\beta}\,\hat{U}_H = \hat{U}_{z,-\frac{\pi}{2}}\,\hat{U}_{y,\beta}\hat{U}_{z,\frac{\pi}{2}} \tag{5-28}$$

– By further application of $\hat{U}_{z,-\frac{\pi}{2}}$ from the right and of $\hat{U}_{z,\frac{\pi}{2}}$ from the left, we obtain:

$$\hat{U}_{y,\beta} = \hat{U}_{z,\frac{\pi}{2}}\,\hat{U}_H\,\hat{U}_{z,\beta}\,\hat{U}_H\,\hat{U}_{z,-\frac{\pi}{2}} \tag{5-29}$$

as we desired to prove.

- Joining together the former two results, we can claim that **every single-qubit gate can be decomposed into a sequence of phase gates and Hadamard gates**.

$$\hat{U}_{\alpha,\beta,\gamma} = \hat{U}_{z,\alpha}\hat{U}_{y,\beta}\hat{U}_{z,\gamma} = \hat{U}_{z,\alpha+\frac{\pi}{2}}\,\hat{U}_H\,\hat{U}_{z,\beta}\,\hat{U}_H\,\hat{U}_{z,\gamma-\frac{\pi}{2}} \tag{5-30}$$

This result further simplifies the requirements for obtaining a physical quantum computer that operates arbitrarily on single qubits. It only needs to be able to perform phase gates and Hadamard gates; placing these gates in a suitable sequence, any single-qubit gate can be realized.

- As a matter of fact, suitable sequences of single-qubit gates can be exploited to perform arbitrary measurements on a qubit, when a single type of measurement is available in the physical platform. In detail we prove the following. **A measurement of an arbitrary observable $\hat{\sigma}_{\vec{u}}$ can be achieved by a sequence of single-qubit gates and a measurement of $\hat{\sigma}_z$.** The single-qubit gates are performing effectively a basis change, which is placed before the measurement in the computational basis of $\hat{\sigma}_z$.

Proof.

– We know that:

$$\hat{\sigma}_{\vec{u}} = |0_{\vec{u}}\rangle\langle 0_{\vec{u}}| - |1_{\vec{u}}\rangle\langle 1_{\vec{u}}| \equiv \begin{bmatrix} \cos\theta & e^{-\iota\phi}\sin\theta \\ e^{+\iota\phi}\sin\theta & -\cos\theta \end{bmatrix}$$

$$\hat{\sigma}_z = |0\rangle\langle 0| - |1\rangle\langle 1| \equiv \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

with $\vec{u} = \begin{bmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{bmatrix}$. Thus we can write:

$$\hat{\sigma}_{\vec{u}} = \Big(|0_{\vec{u}}\rangle\langle 0| + |1_{\vec{u}}\rangle\langle 1|\Big)\Big(|0\rangle\langle 0| - |1\rangle\langle 1|\Big)\Big(|0\rangle\langle 0_{\vec{u}}| + |1\rangle\langle 1_{\vec{u}}|\Big) = \hat{U}_R^\dagger\hat{\sigma}_z\hat{U}_R \tag{5-31}$$

where the operator $\hat{U}_R = |0\rangle\langle 0_{\vec{u}}| + |1\rangle\langle 1_{\vec{u}}|$ is performing the basis change.

– A decomposition for $\hat{U}_R$ can be found by reminding the definitions of $|0_{\vec{u}}\rangle$ and $|1_{\vec{u}}\rangle$, whose Hermitian conjugates are:

$$\langle 0_{\vec{u}}| = \cos\frac{\theta}{2}\langle 0| + e^{-\iota\phi}\sin\frac{\theta}{2}\langle 1| \qquad \langle 1_{\vec{u}}| = \sin\frac{\theta}{2}\langle 0| - e^{-\iota\phi}\cos\frac{\theta}{2}\langle 1| \tag{5-32}$$

Hence:

$$\hat{U}_R = |0\rangle\langle 0_{\vec{u}}| + |1\rangle\langle 1_{\vec{u}}| = \cos\frac{\theta}{2}|0\rangle\langle 0| + e^{-\iota\phi}\sin\frac{\theta}{2}|0\rangle\langle 1| + \sin\frac{\theta}{2}|1\rangle\langle 0| - e^{-\iota\phi}\cos\frac{\theta}{2}|1\rangle\langle 1| \tag{5-33}$$

which written in matrix form is:

$$U_R = \begin{bmatrix} \cos\frac{\theta}{2} & e^{-\iota\phi}\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & -e^{-\iota\phi}\cos\frac{\theta}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & e^{-\iota\phi} \end{bmatrix} \begin{bmatrix} \cos\frac{\theta}{2} & e^{-\iota\phi}\sin\frac{\theta}{2} \\ e^{+\iota\phi}\sin\frac{\theta}{2} & -\cos\frac{\theta}{2} \end{bmatrix} \tag{5-34}$$

We recognize:

$$\hat{U}_{z,-\phi} \equiv \begin{bmatrix} 1 & 0 \\ 0 & e^{-\iota\phi} \end{bmatrix} \qquad \hat{\sigma}_{\vec{v}} = \begin{bmatrix} \cos\frac{\theta}{2} & e^{-\iota\phi}\sin\frac{\theta}{2} \\ e^{+\iota\phi}\sin\frac{\theta}{2} & -\cos\frac{\theta}{2} \end{bmatrix} \tag{5-35}$$

where the vector $\vec{v}$ is defined by the angles $\theta/2$ and ϕ, i.e. $\vec{v} = \begin{bmatrix} \sin\frac{\theta}{2}\cos\phi \\ \sin\frac{\theta}{2}\sin\phi \\ \cos\frac{\theta}{2} \end{bmatrix}$. We may indicate this operator as $\hat{\sigma}_{\frac{\theta}{2},\phi}$.
In conclusion, the basis change operator is:

$$\hat{U}_R = \hat{U}_{z,-\phi}\hat{\sigma}_{\frac{\theta}{2},\phi} \tag{5-36}$$

– The operator $\hat{\sigma}_{\vec{u}}$ now reads:

$$\hat{\sigma}_{\vec{u}} = \hat{U}_R^\dagger \hat{\sigma}_z \hat{U}_R = \hat{\sigma}^\dagger_{\frac{\theta}{2},\phi}\hat{U}^\dagger_{z,-\phi}\hat{\sigma}_z\hat{U}_{z,-\phi}\hat{\sigma}_{\frac{\theta}{2},\phi} \tag{5-37}$$

But $\hat{U}_{z,-\phi}$, $\hat{U}^\dagger_{z,-\phi}$ and $\hat{\sigma}_z$ all commute, as they are rotations about the same z axis in the Bloch sphere. Therefore, one can write more simply:

$$\hat{\sigma}_{\vec{u}} = \hat{\sigma}^\dagger_{\frac{\theta}{2},\phi}\hat{\sigma}_z\hat{\sigma}_{\frac{\theta}{2},\phi} \tag{5-38}$$

– Let us evaluate the expectation value of $\hat{\sigma}_{\vec{u}}$ on a generic qubit state:

$$\langle\hat{\sigma}_{\vec{u}}\rangle = \langle\psi|\,\hat{\sigma}^\dagger_{\frac{\theta}{2},\phi}\hat{\sigma}_z\hat{\sigma}_{\frac{\theta}{2},\phi}\,|\psi\rangle = \langle\psi'|\,\hat{\sigma}_z\,|\psi'\rangle \tag{5-39}$$

where $|\psi'\rangle = \hat{\sigma}_{\frac{\theta}{2},\phi}|\psi\rangle$. The measurement of $\hat{\sigma}_{\vec{u}}$ on $|\psi\rangle$ thus equals the measurement of $\hat{\sigma}_z$ on $|\psi'\rangle$. Put in other words, conducting the measurement of $\hat{\sigma}_{\vec{u}}$ on a generic qubit is equivalent to applying first the single-qubit gate $\hat{\sigma}_{\frac{\theta}{2},\phi}$, and then to measuring $\hat{\sigma}_z$.

This result means that a physical quantum computer can perform a qubit measurement in any basis, if it is able to measure one observable, which defines the computational basis, and is able to perform arbitrary qubit rotations.